

Databricks for Pharmacy Analytics

BCBS Tennessee — A Hands-On Workshop for SAS Users

Today's Agenda

- Foundations: SAS to Databricks, Lakehouse, Unity Catalog, Medallion
- Build the pharmacy data pipeline (Bronze, Silver, Gold)
- Deliver 7 pharmacy analytics use cases your team owns
- AI Assistant + AI/BI Genie for ad-hoc work
- Next steps for your transition

Foundations

Why Databricks for Pharmacy Analytics

- Vendor-feed scale: PBM extracts, NDC tables, eligibility — billions of rows in seconds
- Recurring + ad-hoc in one governed environment
- Faster turnaround means more time for trusted-advisor analysis
- Your SAS knowledge transfers — about 90% of PROC SQL syntax just works

SAS to Databricks – Quick Reference

What you do in SAS	How you do it in Databricks	Difficulty
PROC SQL	SQL queries (almost identical)	Easy
DATA step / IF-THEN	SELECT with CASE WHEN	Easy
LIBNAME	catalog.schema.table (Unity Catalog)	Easy
%MACRO / %LET	Notebook widgets, parameterized SQL	Moderate
PROC TRANSPOSE	PIVOT	Moderate
PROC APPEND / SET	UNION ALL	Easy
PROC FORMAT	CASE WHEN or lookup join	Easy
PROC UNIVARIATE	PERCENTILE_CONT	Easy
PROC EXPAND (trending)	Window functions (LAG, LEAD)	Moderate

Lakehouse + Unity Catalog

Lakehouse

Super-powered SAS library

All pharmacy data in one place

Query billions of rows in seconds

Full audit trail for legislative analyses

Unity Catalog (your LIBNAMEs)

catalog.schema.table

Governed, column-level security for PHI/PII

Lineage and audit out of the box

Searchable — stakeholders self-serve

Medallion Architecture

Bronze

Raw vendor feeds as-is
PBM extracts, eligibility
files
Vendor-feed audit and
anomaly catch

Silver

Cleaned, deduped, typed
Governed source-of-truth
Replaces work.* SAS
datasets

Gold

Pharmacy analytics
deliverables
PMPM, channel mix, UM,
savings
Feeds dashboards and
ad-hocs

Pharmacy Analytics in Gold

Use Cases 1-2: Cost & Trending

1. Drug Spend by Channel

Spend by Retail / Mail Order / Specialty

State breakdown

Maps providers.specialty to channel

Monthly recurring deliverable

2. PMPM Trending

Drug spend, plan paid, claim count

MoM and YoY growth (LAG/LEAD)

3-month moving average smoothing

Feeds rate development

Use Cases 3-4: UM & Risk

3. Utilization Management

Volume by decision status

Monthly approval rate by LOB

Turnaround pattern (proxy in workshop)

For UM team operations review

4. High-Cost / Polypharmacy

95th/90th/75th percentile tiers

Distinct drugs per member

Clinical-program candidate flags

Care management input

Use Cases 5-7: Mix, Quality, Impact

5. Drug Type Mix

Brand / Generic /
Specialty

Proxy from cost bands

Swap NDC join in
production

6. Pharmacy Data Quality

Completeness, accuracy,
duplicates

Monthly close gate

PBM-feed anomaly
detection

7. Program Savings + Impact

Pre/post savings analysis

Channel-mix shift

Legislative cohort
diff-in-diff

AI Assistant + AI/BI Genie

Speed Up Ad-Hoc and Stakeholder Work

AI Assistant

Ask in plain English

SAS to Databricks code conversion

Debug, optimize, explain

Great for ad-hoc requests

AI/BI + Genie

Self-serve dashboards for stakeholders

Natural-language questions on
pharmacy data

Reduces our ad-hoc inbox

Lineage-aware answers

Next Steps for the Team

- Pick one recurring SAS report and port it to Gold this week
- Add the DQ checks to your monthly close
- Parameterize PMPM by LOB using widgets (your %LET equivalents)
- Try Genie on the Silver layer for ad-hoc requests
- Review the SAS-to-Databricks reference cell when you hit a new pattern



databricks